PubH7485_homework1_key

2024-09-16

For dataset 'nhefs'

Question 1 (10 points)

The following table summarizes the differences in covariates between the Control group and the Treatment group:

Table 1: Summary Statistics Stratified by if Quit smoking between 1st Questionnaire and 1982

	level	No	Yes	p	test	SMD
n		1201	428			
Sex (%)	Male	562 (46.8)	237 (55.4)	0.003		0.172
	Female	639 (53.2)	191 (44.6)			
Age in 1971 (mean (SD))		42.92 (11.89)	46.70 (12.52)	<0.001		0.309
Race in 1971 (%)	White	1024 (85.3)	390 (91.1)	0.003		0.182
	Black or Other	177 (14.7)	38 (8.9)			
Total family income in 1971 (mean (SD))		17.90 (2.71)	18.07 (2.52)	0.277		0.064
Marital status in 1971 (%)	Under 17	0 (0.0)	0 (0.0)	NaN		0.130
	Married	933 (77.7)	346 (80.8)			
	Widowed	70 (5.8)	27 (6.3)			
	Never married	76 (6.3)	20 (4.7)			
	Divorced	77 (6.4)	22 (5.1)			
	Separated	45 (3.7)	12 (2.8)			
	Unknown	0 (0.0)	1 (0.2)			
Highest grade of regular school ever in 1971 (mean (SD))		11.12 (3.00)	11.17 (3.32)	0.782		0.015
Amount of education in 1971 (%)	8th grade or less	218 (18.2)	93 (21.7)	0.013		0.197
	HS dropout	273 (22.7)	78 (18.2)			
	HS	495 (41.2)	164 (38.3)			
	College dropout	96 (8.0)	30 (7.0)			
	College or more	119 (9.9)	63 (14.7)			
Height in centimeters in 1971 (mean (SD))		168.48 (9.00)	169.48 (9.17)	0.050		0.110
Weight in kilograms in 1971 (mean (SD))		70.49 (15.57)	72.63 (16.08)	0.015		0.135

	level	No	Yes	p	test	SMD
Number of cigarettes smoked per day in 1971 (mean (SD))		21.18 (11.58)	18.79 (12.26)	<0.001		0.200
Diagnosed asthma in 1971 (%)	Never	1149 (95.7)	401 (93.7)	0.132		0.088
	Ever	52 (4.3)	27 (6.3)			
Diagnosed chronic bronchitis/emphysema in 1971 (%)	Never	1101 (91.7)	389 (90.9)	0.690		0.028
	Ever	100 (8.3)	39 (9.1)			
Diagnosed tuberculosis in 1971 (%)	Never	1188 (98.9)	418 (97.7)	0.099		0.097
	Ever	13 (1.1)	10 (2.3)			
Diagnosed heart failure in 1971 (%)	Never	1195 (99.5)	426 (99.5)	1.000		0.005
	Ever	6 (0.5)	2 (0.5)			
Diagnosed high blood pressure in 1971 (%)	Never	524 (43.6)	184 (43.0)	0.260		0.090
	Ever	88 (7.3)	42 (9.8)			
	Missing	589 (49.0)	202 (47.2)			
Diagnosed peptic ulcer in 1971 (%)	Never	1083 (90.2)	377 (88.1)	0.260		0.067
	Ever	118 (9.8)	51 (11.9)			
Diagnosed colitis in 1971 (%)	Never	1156 (96.3)	418 (97.7)	0.218		0.082
	Ever	45 (3.7)	10 (2.3)			
Diagnosed hepatitis in 1971 (%)	Never	1179 (98.2)	422 (98.6)	0.711		0.034
	Ever	22 (1.8)	6 (1.4)			
Diagnosed chronic cough in 1971 (%)	Never	1140 (94.9)	401 (93.7)	0.400		0.053
	Ever	61 (5.1)	27 (6.3)			
Diagnosed hay fever in 1971 (%)	Never	1096 (91.3)	387 (90.4)	0.673		0.029
	Ever	105 (8.7)	41 (9.6)			
Diagnosed diabetes in 1971 (%)	Never	600 (50.0)	224 (52.3)	0.446		0.075
	Ever	12 (1.0)	2 (0.5)			
	Missing	589 (49.0)	202 (47.2)			
Diagnosed polio in 1971 (%)	Never	1186 (98.8)	420 (98.1)	0.487		0.050
	Ever	15 (1.2)	8 (1.9)			
Diagnosed malignant tumor/growth in 1971 (%)	Never	1172 (97.6)	419 (97.9)	0.857		0.021
	Ever	29 (2.4)	9 (2.1)			
Diagnosed nervous breakdown in 1971 (%)	Never	1163 (96.8)	419 (97.9)	0.338		0.066
	Ever	38 (3.2)	9 (2.1)			
Have you had 1 drink past year? (%)	Never	146 (12.2)	61 (14.3)	0.513		0.064
	Ever	1051 (87.5)	366 (85.5)			
	Missing	4 (0.3)	1 (0.2)			
How often do you drink? (%)	Almost every day	247 (20.6)	89 (20.8)	0.526		0.116
	2-3 times/week	179 (14.9)	52 (12.1)			

	level	No	Yes	p	test	SMD
	1-4 times/month	365 (30.4)	141 (32.9)			
	< 12 times/year	260 (21.6)	84 (19.6)			
	No alcohol last year	146 (12.2)	61 (14.3)			
	Unknown	4 (0.3)	1 (0.2)			
Which do you most frequently drink? (%)	Beer	429 (35.7)	140 (32.7)	0.734		0.064
	Wine	96 (8.0)	36 (8.4)			
	Liquor	372 (31.0)	140 (32.7)			
	Other/Unknown	304 (25.3)	112 (26.2)			
When you drink, how much do you drink? (mean (SD))		3.35 (3.13)	3.09 (2.53)	0.183		0.091
Do you eat dirt or clay, starch or other non-standard food? (%)	Never	606 (50.5)	225 (52.6)	0.601		0.059
	Ever	6 (0.5)	1 (0.2)			
	Missing	589 (49.0)	202 (47.2)			
Use headache medication in 1971 (%)	Never	434 (36.1)	169 (39.5)	0.240		0.069
	Ever	767 (63.9)	259 (60.5)			
Use other pains medication in 1971 (%)	Never	905 (75.4)	323 (75.5)	1.000		0.003
	Ever	296 (24.6)	105 (24.5)			
Use weak heart medication in 1971 (%)	Never	1173 (97.7)	420 (98.1)	0.714		0.032
	Ever	28 (2.3)	8 (1.9)			
Use allergies medication in 1971 (%)	Never	1129 (94.0)	399 (93.2)	0.647		0.032
	Ever	72 (6.0)	29 (6.8)			
Use nerves medication in 1971 (%)	Never	1021 (85.0)	373 (87.1)	0.317		0.062
	Ever	180 (15.0)	55 (12.9)			
Use lack of pep medication in 1971 (%)	Never	1136 (94.6)	410 (95.8)	0.397		0.056
	Ever	65 (5.4)	18 (4.2)			
Use high blood pressure medication in 1971 (%)	Never	578 (48.1)	204 (47.7)	0.077		0.119
	Ever	34 (2.8)	22 (5.1)			
	Missing	589 (49.0)	202 (47.2)			
Use bowel trouble medication in 1971 (%)	Never	536 (44.6)	198 (46.3)	0.806		0.037
	Ever	76 (6.3)	28 (6.5)			
	Missing	589 (49.0)	202 (47.2)			
Use weight loss medication in 1971 (%)	Never	1167 (97.2)	420 (98.1)	0.368		0.064
	Ever	34 (2.8)	8 (1.9)			
Use infection medication in 1971 (%)	Never	1019 (84.8)	369 (86.2)	0.545		0.039
	Ever	182 (15.2)	59 (13.8)			
In your usual day, how active are you? (%)	Very active	547 (45.5)	182 (42.5)	0.427		0.073
	Moderately active	540 (45.0)	198 (46.3)			
	Inactive	114 (9.5)	48 (11.2)			
In recreation, how much exercise? (%)	Little or no exercise	247 (20.6)	70 (16.4)	0.150		0.112

	level	No	Yes	p	test	SMD
	Moderate exercise	496 (41.3)	181 (42.3)			
	Much exercise	458 (38.1)	177 (41.4)			
Birth control pills past 6 months? (%)	No	515 (42.9)	153 (35.7)	0.012		0.168
	Yes	118 (9.8)	37 (8.6)			
	Missing	568 (47.3)	238 (55.6)			
Total number of pregnancies? (mean (SD))		3.67 (2.20)	3.78 (2.22)	0.566		0.050
Serum cholesterol (mg/100ml) in 1971 (mean (SD))		218.90 (45.13)	222.96 (46.22)	0.114		0.089
AVG TOBACCO PRICE IN STATE OF RESIDENCE 1971 (US$2008) (mean (SD))		2.14 (0.23)	2.14 (0.23)	0.776		0.017
TOBACCO TAX IN STATE OF RESIDENCE 1971 (US$2008) (mean (SD))		1.06 (0.21)	1.06 (0.22)	0.892		0.008

Question 2 (15 points, 5 for each outcome)

Using a confidence level of 95%, the unadjusted average treatment effect for the outcome of weight change between 1971 to 1982 was evaluated as **2.54, with a standard error of 0.47 and a confidence interval of (1.61, 3.47)**.

Using a confidence level of 95%, the unadjusted average treatment effect for the outcome of SBP in 1982 was evaluated as **3.99, with a standard error of 1.09 and a confidence interval of (1.85, 6.13)**.

Using a confidence level of 95%, the unadjusted average treatment effect for the outcome of DBP in 1982 was evaluated as **1.53, with a standard error of 0.61 and a confidence interval of (0.34, 2.71)**.

Table 2: Outcome

	ATE	Standard Error	95% CI
Weight Change	2.54	0.47	1.61, 3.47
SBP	3.99	1.09	1.85, 6.13
DBP	1.53	0.61	0.34, 2.71

For dataset 'black_politicians'

Question 3 (10 points)

The following table summarizes the differences in covariates between the Control group and the Treatment group:

Table 3: Summary Statistics Stratified by if Legislator Receiving Email is Black

	level	No	Yes	p	test	SMD
n		5229	364			
Email is from out of district (%)	No	2629 (50.3)	185 (50.8)	0.883		0.011

	level	No	Yes	p	test	SMD
	Yes	2600 (49.7)	179 (49.2)			
District population (mean (SD))		8.42 (10.86)	10.43 (9.52)	0.001		0.197
District median household income (mean (SD))		4.43 (1.40)	3.33 (1.14)	<0.001		0.867
District median household income among Black people (mean (SD))		1.57 (1.02)	1.37 (0.43)	<0.001		0.259
District median household income among White people (mean (SD))		2.33 (0.77)	2.22 (0.65)	0.009		0.152
Percentage of district that is Black (mean (SD))		0.06 (0.10)	0.52 (0.20)	<0.001		2.869
State's Squire index (mean (SD))		0.19 (0.12)	0.21 (0.12)	0.001		0.175
Legislator receiving email is neither Black nor White (%)	No	4965 (95.0)	364 (100.0)	<0.001		0.326
	Yes	264 (5.0)	0 (0.0)			
Percentage of district that is urban (mean (SD))		0.67 (0.32)	0.85 (0.25)	<0.001		0.630
Legislator receiving email is a senator (%)	No	3839 (73.4)	278 (76.4)	0.240		0.068
	Yes	1390 (26.6)	86 (23.6)			
Legislator receiving email is in the Democratic party (%)	No	2611 (49.9)	8 (2.2)	<0.001		1.296
	Yes	2618 (50.1)	356 (97.8)			
Legislator receiving email is in the Southern United States (%)	No	3920 (75.0)	168 (46.2)	<0.001		0.617
	Yes	1309 (25.0)	196 (53.8)			

Question 4 (5 points)

Using a confidence level of 95%, the unadjusted average treatment effect for the binary outcome of whether legistor reponded to the email was evaluated as **-0.032, with a standard error of 0.027 and a confidence interval of (-0.084, 0.020)**.

Table 4: Outcome

	ATE	Standard Error	95% CI
repond	-0.032	0.027	-0.084, 0.020

Appendix - R Code

For the dataset 'nhefs':

1. Clean the dataset, remove unused variables and give others clear labels; factor the categorical variables and give clear labels for each level.

```r
# treatment
nhefs$qsmk
# outcomes of interest
nhefs$wt82_71
nhefs$sbp
nhefs$dbp

df <- nhefs[, setdiff(names(nhefs), c('dadth', 'death', 'hightax82', 'modth', 'price71_82',
                                      'price82', 'seqn', 'yrdth', 'wt82', 'tax71_82',
                                      'tax82', 'smokeyrs', 'smkintensity82_71', 'birthplace','id','cens


# Convert categorical variables to factors and set levels for them
df$active <- factor(df$active, levels = c(0, 1, 2), labels = c("Very active", "Moderately active", "Inac
df$alcoholfreq <- factor(df$alcoholfreq, levels = c(0, 1, 2, 3, 4, 5), labels = c("Almost every day", "
df$alcoholpy <- factor(df$alcoholpy, levels = c(0, 1, 2), labels = c("Never", "Ever", "Missing"))
df$alcoholtype <- factor(df$alcoholtype, levels = c(1, 2, 3, 4), labels = c("Beer", "Wine", "Liquor", "
df$allergies <- factor(df$allergies, levels = c(0, 1), labels = c("Never", "Ever"))
df$asthma <- factor(df$asthma, levels = c(0, 1), labels = c("Never", "Ever"))
df$birthcontrol <- factor(df$birthcontrol, levels = c(0, 1, 2), labels = c("No", "Yes", "Missing"))
df$boweltrouble <- factor(df$boweltrouble, levels = c(0, 1, 2), labels = c("Never", "Ever", "Missing"))
df$bronch <- factor(df$bronch, levels = c(0, 1), labels = c("Never", "Ever"))
df$chroniccough <- factor(df$chroniccough, levels = c(0, 1), labels = c("Never", "Ever"))
df$colitis <- factor(df$colitis, levels = c(0, 1), labels = c("Never", "Ever"))
df$diabetes <- factor(df$diabetes, levels = c(0, 1, 2), labels = c("Never", "Ever", "Missing"))
df$education <- factor(df$education, levels = c(1, 2, 3, 4, 5), labels = c("8th grade or less", "HS drop
df$exercise <- factor(df$exercise, levels = c(0, 1, 2), labels = c("Little or no exercise", "Moderate ex
df$hayfever <- factor(df$hayfever, levels = c(0, 1), labels = c("Never", "Ever"))
df$hbp <- factor(df$hbp, levels = c(0, 1, 2), labels = c("Never", "Ever", "Missing"))
df$hbpmed <- factor(df$hbpmed, levels = c(0, 1, 2), labels = c("Never", "Ever", "Missing"))
df$headache <- factor(df$headache, levels = c(0, 1), labels = c("Never", "Ever"))
df$hepatitis <- factor(df$hepatitis, levels = c(0, 1), labels = c("Never", "Ever"))
df$hf <- factor(df$hf, levels = c(0, 1), labels = c("Never", "Ever"))
df$marital <- factor(df$marital, levels = c(1, 2, 3, 4, 5, 6, 8), labels = c("Under 17", "Married", "Wid
df$nerves <- factor(df$nerves, levels = c(0, 1), labels = c("Never", "Ever"))
df$nervousbreak <- factor(df$nervousbreak, levels = c(0, 1), labels = c("Never", "Ever"))
df$otherpain <- factor(df$otherpain, levels = c(0, 1), labels = c("Never", "Ever"))
df$pepticulcer <- factor(df$pepticulcer, levels = c(0, 1), labels = c("Never", "Ever"))
df$pica <- factor(df$pica, levels = c(0, 1, 2), labels = c("Never", "Ever", "Missing"))
df$polio <- factor(df$polio, levels = c(0, 1), labels = c("Never", "Ever"))
df$qsmk <- factor(df$qsmk, levels = c(0, 1), labels = c("No", "Yes"))
df$race <- factor(df$race, levels = c(0, 1), labels = c("White", "Black or Other"))
df$sex <- factor(df$sex, levels = c(0, 1), labels = c("Male", "Female"))
df$tb <- factor(df$tb, levels = c(0, 1), labels = c("Never", "Ever"))
df$tumor <- factor(df$tumor, levels = c(0, 1), labels = c("Never", "Ever"))
df$weakheart <- factor(df$weakheart, levels = c(0, 1), labels = c("Never", "Ever"))
df$wtloss <- factor(df$wtloss, levels = c(0, 1), labels = c("Never", "Ever"))
df$lackpep <- factor(df$lackpep, levels = c(0, 1), labels = c("Never", "Ever"))
df$infection <- factor(df$infection, levels = c(0, 1), labels = c("Never", "Ever"))

df <- set_variable_labels(df,
  active = "In your usual day, how active are you?",
  age = "Age in 1971",
```

```r
  alcoholfreq = "How often do you drink?",
  alcoholhowmuch = "When you drink, how much do you drink?",
  alcoholpy = "Have you had 1 drink past year?",
  alcoholtype = "Which do you most frequently drink?",
  allergies = "Use allergies medication in 1971",
  asthma = "Diagnosed asthma in 1971",
  birthcontrol = "Birth control pills past 6 months?",
  boweltrouble = "Use bowel trouble medication in 1971",
  bronch = "Diagnosed chronic bronchitis/emphysema in 1971",
  cholesterol = "Serum cholesterol (mg/100ml) in 1971",
  chroniccough = "Diagnosed chronic cough in 1971",
  colitis = "Diagnosed colitis in 1971",
  dbp = "Diastolic blood pressure in 1982",
  diabetes = "Diagnosed diabetes in 1971",
  education = "Amount of education in 1971",
  exercise = "In recreation, how much exercise?",
  hayfever = "Diagnosed hay fever in 1971",
  hbp = "Diagnosed high blood pressure in 1971",
  hbpmed = "Use high blood pressure medication in 1971",
  headache = "Use headache medication in 1971",
  hepatitis = "Diagnosed hepatitis in 1971",
  hf = "Diagnosed heart failure in 1971",
  ht = "Height in centimeters in 1971",
  income = "Total family income in 1971",
  infection = "Use infection medication in 1971",
  lackpep = "Use lack of pep medication in 1971",
  marital = "Marital status in 1971",
  nerves = "Use nerves medication in 1971",
  nervousbreak = "Diagnosed nervous breakdown in 1971",
  otherpain = "Use other pains medication in 1971",
  pepticulcer = "Diagnosed peptic ulcer in 1971",
  pica = "Do you eat dirt or clay, starch or other non-standard food?",
  polio = "Diagnosed polio in 1971",
  pregnancies = "Total number of pregnancies?",
  qsmk = "Quit smoking between 1st questionnaire and 1982",
  race = "Race in 1971",
  sbp = "Systolic blood pressure in 1982",
  school = "Highest grade of regular school ever in 1971",
  sex = "Sex",
  smokeintensity = "Number of cigarettes smoked per day in 1971",
  tb = "Diagnosed tuberculosis in 1971",
  tumor = "Diagnosed malignant tumor/growth in 1971",
  weakheart = "Use weak heart medication in 1971",
  wt71 = "Weight in kilograms in 1971",
  wt82_71 = "Weight change in kilograms",
  wtloss = "Use weight loss medication in 1971"
)

vars_to_include1 <- setdiff(names(df), c('qsmk','wt82_71','sbp','dbp'))
```

2. Create table one

```r
t1 <- CreateTableOne(var = vars_to_include1, data = df, strata = 'qsmk')
t11 <- print(t1, smd = TRUE, showAllLevels = TRUE, varLabels = TRUE)
kable(t11, caption = "Summary Statistics Stratified by if Quit smoking between 1st Questionnaire and 198
```

3. Obtain the unadjusted ATE, se and confidence interval for each variable

```r
num_t1 <- table(df$qsmk)[2]
num_c1 <- table(df$qsmk)[1]

## for weight change between 1971 and 1982 (wt82_71)
wc_t <- df$wt82_71[df$qsmk == 'Yes']
wc_c <- df$wt82_71[df$qsmk == 'No']
et_wc <- mean(wc_t, na.rm = T)
ec_wc <- mean(wc_c, na.rm = T)
# Unadjusted ATE
ate_wc <- et_wc - ec_wc
print(ate_wc, digits = 3)
# Standard Error
se_wc <- sqrt(var(wc_t, na.rm = T)/num_t1 + var(wc_c, na.rm = T)/num_c1)
print(se_wc, digits = 3)
# Confidence Interval
conf_int_wc <- ate_wc + c(-1, 1)*qnorm(0.975)*se_wc
print(conf_int_wc, digits = 3)

## for systolic (sbp) in 1982
sbp_t <- df$sbp[df$qsmk == 'Yes']
sbp_c <- df$sbp[df$qsmk == 'No']
et_sbp <- mean(sbp_t, na.rm = T)
ec_sbp <- mean(sbp_c, na.rm = T)
# Unadjusted ATE
ate_sbp <- et_sbp - ec_sbp
print(ate_sbp, digits = 3)
# Standard Error
se_sbp <- sqrt(var(sbp_t, na.rm = T)/num_t1 + var(sbp_c, na.rm = T)/num_c1)
print(se_sbp, digits = 3)
# Confidence Interval
conf_int_sbp <- ate_sbp + c(-1, 1)*qnorm(0.975)*se_sbp
print(conf_int_sbp, digits = 3)

## for diastolic blood pressure (dbp) in 1982
dbp_t <- df$dbp[df$qsmk == 'Yes']
dbp_c <- df$dbp[df$qsmk == 'No']
et_dbp <- mean(dbp_t, na.rm = T)
ec_dbp <- mean(dbp_c, na.rm = T)
# Unadjusted ATE
ate_dbp <- et_dbp - ec_dbp
print(ate_dbp, digits = 3)
# Standard Error
se_dbp <- sqrt(var(dbp_t, na.rm = T)/num_t1 + var(dbp_c, na.rm = T)/num_c1)
print(se_dbp, digits = 3)
# Confidence Interval
conf_int_dbp <- ate_dbp + c(-1, 1)*qnorm(0.975)*se_dbp
print(conf_int_dbp, digits = 3)
```

For the dataset 'black_politicians':

1. Clean the dataset, remove unused variables and give others clear labels; factor the categorical variables and give clear labels for each level.

```
# data("black_politicians")
head(black_politicians)
# treatment
black_politicians$leg_black
# outcome
black_politicians$responded

df2 <- black_politicians

df2$leg_black <- as.factor(df2$leg_black)
levels(df2$leg_black) <- c("No", "Yes")
df2$treat_out <- as.factor(df2$treat_out)
levels(df2$treat_out) <- c("No", "Yes")
df2$nonblacknonwhite <- as.factor(df2$nonblacknonwhite)
levels(df2$nonblacknonwhite) <- c("No", "Yes")
df2$leg_senator <- as.factor(df2$leg_senator)
levels(df2$leg_senator) <- c("No", "Yes")
df2$leg_democrat <- as.factor(df2$leg_democrat)
levels(df2$leg_democrat) <- c("No", "Yes")
df2$south <- as.factor(df2$south)
levels(df2$south) <- c("No", "Yes")

df2 <- set_variable_labels(df2, leg_black = "Legislator receiving email is Black",
                           treat_out = "Email is from out of district",
                           responded = "Legislator responded to email",
                           totalpop = "District population",
                           medianhhincom = "District median household income",
                           black_medianhh = "District median household income among Black people",
                           white_medianhh = "District median household income among White people",
                           blackpercent = "Percentage of district that is Black",
                           statessquireindex = "State's Squire index",
                           nonblacknonwhite = "Legislator receiving email is neither Black nor White",
                           urbanpercent = "Percentage of district that is urban",
                           leg_senator = "Legislator receiving email is a senator",
                           leg_democrat = "Legislator receiving email is in the Democratic party",
                           south = "Legislator receiving email is in the Southern United States")

vars_to_include2 <- setdiff(names(df2), c('responded','leg_black'))
```

2. Create table one

```
t2 <- CreateTableOne(var = vars_to_include2, data = df2, strata = 'leg_black')
t22 <- print(t2, smd = TRUE, showAllLevels = TRUE,  varLabels = TRUE)
kable(t22, caption = "Summary Statistics Stratified by if Legislator Receiving Email is Black")
```

3. Obtain the unadjusted ATE, se and confidence interval for each variable

```r
et_responded <- mean(df2$responded[df2$leg_black=='Yes'])
ec_responded <- mean(df2$responded[df2$leg_black=='No'])
num_t <- table(df2$leg_black)[2]
num_c <- table(df2$leg_black)[1]
# Unadjusted ATE
ate_responded <- et_responded - ec_responded
print(ate_responded, digits = 3)
# Standard Error
se_responded <- sqrt(et_responded*(1-et_responded)/num_t + ec_responded*(1-ec_responded)/num_c)
print(se_responded, digits = 3)
# Confidence Interval
conf_int_responded <- ate_responded + c(-1, 1)*qnorm(0.975)*se_responded
print(conf_int_responded, digits = 3)
```